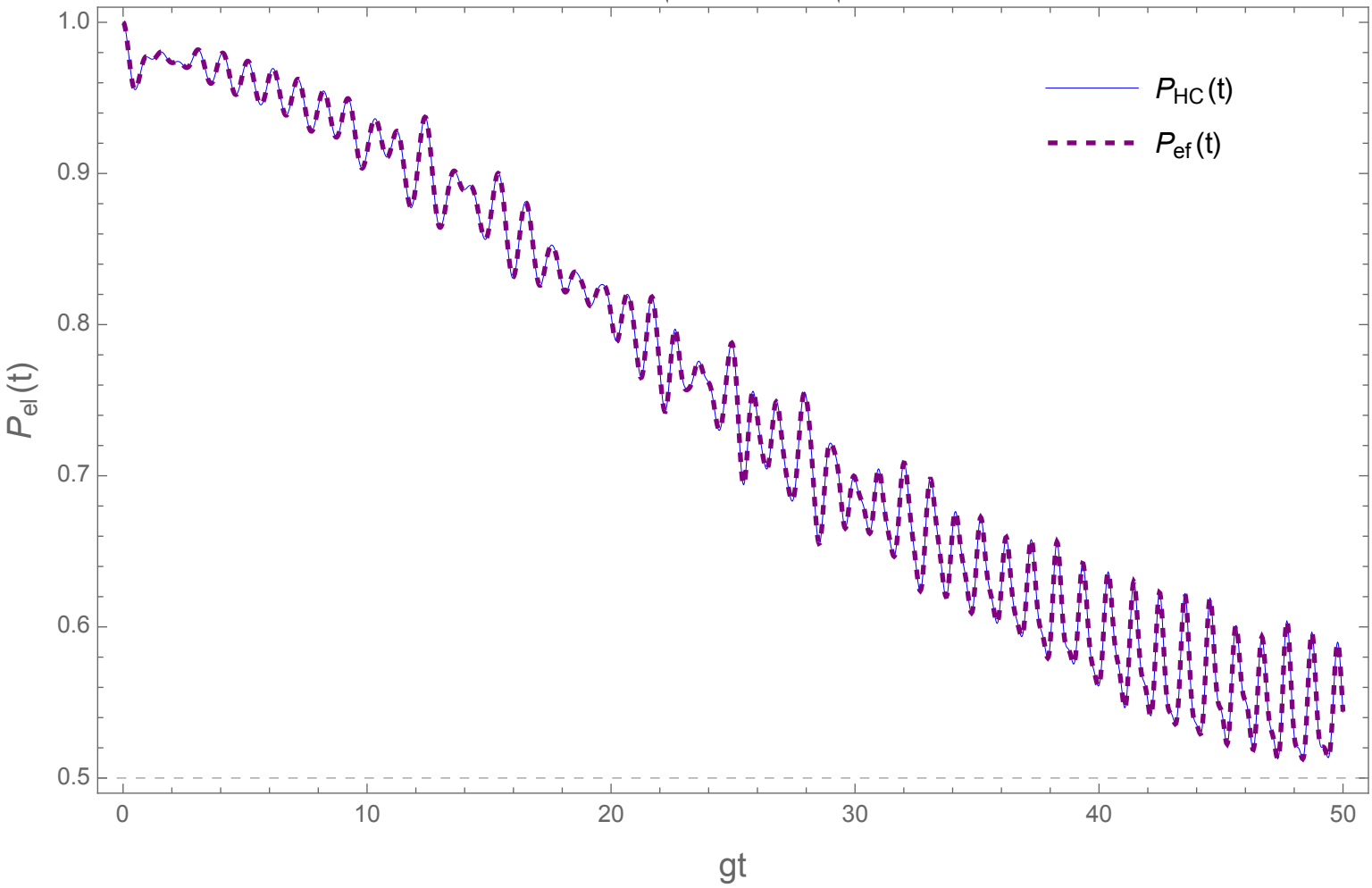$$c_e = \frac{1}{\sqrt{2}}, \quad c_g = \frac{-1}{\sqrt{2}}$$